

GrounUp Enterprise Specification

Phase 31 - Production Deployment, Cloud Infrastructure, DevOps, CI/CD, Observability, DR & SaaS Operations

Version 1.0.0 | Implementation-ready specification

1. Mission

Implement the production infrastructure and operating model required to deploy GrounUp safely, repeatedly, observably and recoverably as a multi-tenant SaaS platform.

2. Provider Neutrality

Phase 31 defines logical cloud, runtime, networking, data, CI/CD, observability and recovery contracts. Specific vendors remain architecture decisions so the enterprise specification is not tied to an unapproved stack.

3. Infrastructure and Environments

Production infrastructure is versioned through IaC with isolated environments, managed secrets, protected state, policy checks and explicit service ownership.

4. CI/CD and Release Engineering

Source changes become immutable artifacts with tests, SBOM/provenance and release evidence. Promotion reuses the tested artifact rather than rebuilding between environments.

5. Safe Deployment

Production releases use controlled migrations, health gates, canary or blue/green strategies where appropriate, feature flags and tested rollback/recovery paths.

6. Observability and SRE

Critical services emit logs, metrics, traces and health signals; SLOs, alerts, owners, on-call rotations and runbooks turn telemetry into operational action.

7. Scale and Resilience

Capacity planning, autoscaling, performance tests, timeouts, retries, circuit breakers and degraded modes protect availability without hiding dependency failure.

8. Backup and Disaster Recovery

Encrypted backups/PITR are paired with restore validation. Critical services require RTO/RPO targets and exercised dependency-ordered DR plans.

9. SaaS Operations

Maintenance, patching, production access, incident response, support, service ownership and FinOps become governed operational capabilities.

10. Production Readiness Boundary

Phase 31 supplies deployment and operations evidence, but production readiness is only achieved when applicable Phase 20, 22, 28 and 31 release/security/business gates are actually approved.

11. Implementation Order

Architecture ADRs -> IaC/environments -> secrets/config -> CI/artifacts -> CD/migrations -> observability/SLOs -> scalability/resilience -> backup/restore -> DR exercises -> runbooks/on-call/FinOps -> production acceptance.

Production Modules

ID	Module	Layer	Purpose
P31-MOD-001	Platform Control Plane	Platform	Environment, service, deployment and ownership registry
P31-MOD-002	Cloud Provider Adapter	Infrastructure	Provider-neutral interface for cloud-specific modules and services
P31-MOD-003	Network Foundation	Infrastructure	VPC/VNet, subnets, routing, NAT/egress and segmentation
P31-MOD-004	Edge & Ingress Service	Infrastructure	DNS, CDN/WAF/ingress/load balancing and TLS termination
P31-MOD-005	Runtime Platform	Infrastructure	Application/container/worker runtime and scheduling
P31-MOD-006	Database Platform	Data Infrastructure	Managed relational database topology, replicas, backups and maintenance
P31-MOD-007	Cache Platform	Data Infrastructure	Distributed cache, locks and rate/ephemeral state
P31-MOD-008	Queue/Job Platform	Data Infrastructure	Asynchronous queues, jobs, dead-letter and worker scaling
P31-MOD-009	Object Storage Platform	Data Infrastructure	Documents, drawings, exports, backups and object lifecycle
P31-MOD-010	Search Platform	Data Infrastructure	Permission-aware search runtime and index operations
P31-MOD-011	Analytics Platform Runtime	Data Infrastructure	Phase 29 warehouse/lakehouse execution infrastructure
P31-MOD-012	AI Runtime Platform	AI Infrastructure	Phase 27 model/provider/tool worker runtime and scaling
P31-MOD-013	Environment Registry	DevOps	Development/test/staging/production/preview environment definitions
P31-MOD-014	Configuration Registry	DevOps	Versioned non-secret environment/application configuration
P31-MOD-015	Secret Injection Service	Security/DevOps	Managed secret delivery, rotation and workload identity integration
P31-MOD-016	Infrastructure-as-Code Service	DevOps	Versioned infrastructure modules and environment stacks
P31-MOD-017	IaC State Service	DevOps	Protected state/locking/backup and change evidence
P31-MOD-018	Policy-as-Code Service	DevSecOps	Infrastructure/security/release policy checks
P31-MOD-019	Build Pipeline	CI	Compile/build/package/test workflow
P31-MOD-020	Artifact Registry	CI	Immutable versioned application/container/package artifacts
P31-MOD-021	SBOM/Provenance Service	DevSecOps	Software bill of materials, signing and build provenance
P31-MOD-022	Test Pipeline	CI	Unit/integration/contract/security/performance test orchestration
P31-MOD-023	Release Candidate Service	Release	Release candidate assembly, versioning and evidence
P31-MOD-024	Deployment Pipeline	CD	Controlled environment promotion and deployment
P31-MOD-025	Migration Orchestrator	CD/Data	Schema/data migration ordering, validation and rollback/forward-fix controls
P31-MOD-026	Feature Flag Runtime	Release	Safe activation/deactivation independent from deployment
P31-MOD-027	Blue/Green Deployment Controller	Release	Parallel production slot deployment and traffic switch
P31-MOD-028	Canary Deployment Controller	Release	Progressive traffic rollout and automated stop conditions
P31-MOD-029	Rollback Service	Release	Application/config/feature rollback and recovery coordination
P31-MOD-030	Release Approval Service	Governance	Human/system release gate approvals and exceptions
P31-MOD-031	Release Evidence Service	Governance	Build/test/security/migration/deployment evidence bundle
P31-MOD-032	Observability Gateway	SRE	Unified logs, metrics, traces and correlation
P31-MOD-033	Logging Platform	SRE	Structured log ingestion, retention, search and redaction
P31-MOD-034	Metrics Platform	SRE	Service/business/platform metric collection and retention
P31-MOD-035	Tracing Platform	SRE	Distributed trace collection and service dependency visibility
P31-MOD-036	SLO/SLI Service	SRE	Service-level objectives, indicators and error budgets

ID	Module	Layer	Purpose
P31-MOD-037	Alerting Service	SRE	Alert routing, deduplication, escalation and maintenance suppression
P31-MOD-038	On-Call Service	SRE	Ownership, rotations, escalation and acknowledgment integration boundary
P31-MOD-039	Incident Operations Service	SRE	Incident declaration, timeline, coordination, comms and closure
P31-MOD-040	Capacity & Autoscaling Service	SRE	Capacity models, scaling policy and saturation signals
P31-MOD-041	Performance Engineering Service	SRE	Load/performance test baselines and regression detection
P31-MOD-042	Resilience Policy Service	SRE	Timeout/retry/circuit-breaker/bulkhead/idempotency policy
P31-MOD-043	Backup Service	DR	Scheduled backup policy, encryption, retention and evidence
P31-MOD-044	Restore Validation Service	DR	Automated/manual restore verification and reconciliation
P31-MOD-045	Disaster Recovery Orchestrator	DR	Regional/service recovery plan, sequencing and failover evidence
P31-MOD-046	Business Continuity Service	Operations	Manual fallback/degraded operations and communication plans
P31-MOD-047	Operational Runbook Registry	Operations	Versioned service runbooks, owners and procedures
P31-MOD-048	FinOps Service	Operations/Finance	Cloud cost allocation, budgets, anomaly detection and optimization

Candidate SLOs

ID	Service	Indicator	Objective	Status
SLO-API-AVAIL	Core API Availability	Availability	99.9% candidate	Draft - owner approval required
SLO-API-LAT	Core API Latency	p95 request latency	<= approved target	Draft - workload targets required
SLO-EST-CALC	Estimate Calculation Reliability	Successful deterministic calculations	>= approved target	Draft
SLO-DOC-UPLOAD	Document Upload Reliability	Successful accepted uploads	>= approved target	Draft
SLO-AI-COPILOT	AI Copilot Service Availability	Successful governed AI requests excluding policy targets	>= approved target	Draft
SLO-ANALYTICS	Certified Dashboard Freshness	Assets within freshness SLA	>= approved target	Draft
SLO-BILLING	Commercial Entitlement Availability	Successful entitlement resolutions	>= approved target	Draft